



1st class
**Research foundations/
RQ creation/
Finding & analyzing papers**

**When you hear the
word **research**, what
comes to mind?**

You can either open up your
microphones and tell possible
answers or write them down in the
chat

What academic research really IS...

Academic research is a systematic process of generating new knowledge, answering questions, or investigating problems using evidence and critical thinking. It should not be anyhow biased or touched with by your own mood and experiences.

Key characteristics?

Systematic

Evidence-based

Objective

Transparent

Reproducible

What Research is NOT

NOT Research	Research
Copying information	Analyzing information
Using random websites	Using credible evidence
Repeating opinions	Building arguments
ChatGPT summaries only	Independent investigation
"I think..."	"Evidence suggests..."

Yeah, of course research is collecting facts. BUT... It is not just it. You guys must create a thorough understanding throughout the paper. A voice.

The Research process

Question → Introduction → Literature Review →
Methodology(different for empirical papers and different for lit.
review papers)
→ Results → Analysis(discussion) → Conclusion → Publication
(optional)

Question - **You as the researcher identify a specific problem or issue you want to investigate**

Introduction - **The study explains the topic and why it is important/research gaps**

Literature Review - **You will need to examine what previous studies have already discovered about the topic**

Method - **The study describes how data or information was collected and analyzed**

Analysis - **You interpret the evidence and identify important findings or patterns**

Conclusion - **The study summarizes the results and explains what they mean**

Publication - **The research is shared with others through journals, conferences, or academic platforms**



**A good day starts
with coffee...**

A good research paper starts with a good question!

Topic vs Research question

Topic is ----
a broad subject area.

Examples can be anything

Climate change

Social media

Artificial intelligence

Education inequality

Research Question though...
is a very specific question you
investigate the whole time while
writing the research paper.

Look at some of the examples ---

**"How does TikTok political content
influence political polarization
among teenagers?"**

**"How has AI changed classroom
learning in high schools?"**

How do we turn the topics into the research questions?

Let's take a look at the example

Potential topic could be "Social media"

Now, what would an uneducated researcher do? Right. Create a weak question like ----

"How does social media affect people?"

It is too broad and vague. You know why? Because it is impossible to answer fully. Social media affects people in 10000000+ different ways. Sleeping, spending money, shifting eyesight, having bad terms with parents and other million things.

A better question here would be

"How does TikTok political content influence political polarization among teenagers in the U.S.?"

Why better? It mentions specific platform and specific effect. It also has a specific population. Much more researchable

Formula for strong research questions

How/Why does [factor] affect [outcome] among [group/context]?

More examples for you guys to understand ----

- 1. How does sleep deprivation affect academic performance among high school students?
- 2. Why do young voters trust social media news sources?
- 3. How does AI-assisted learning influence student productivity?

Characteristics you must have	What specifically that means?
Focused	Not too broad
Clear	Easy to understand
Researchable	Evidence can be collected
Specific	Clearly defined variables/groups
Significant	Actually matters



How do I understand that my RQ is strong enough?

Weak vs strong questions

Weak Question	Why Weak?
Is social media bad?	Opinion-based
What is climate change?	Too descriptive
Why is education important?	Too broad

Alright. Better ones?

Strong Question	Why Strong?
How does Instagram usage affect self-esteem among teenagers?	Focused + measurable
How do urban heatwaves impact public health in Tashkent?	Specific context
How does AI use influence student writing habits?	Clear variables



Research Question Red Flags

Dangerous Questions are

1. too broad
2. impossible to measure
3. purely opinion-based
4. emotionally loaded
5. requires inaccessible data (we do not work with NASA labs)

Example here ----

“Why is capitalism evil?”

Problems? It has really biased wording and is not academically neutral. Some people like capitalism. Most people in USA)

Literature Review vs Empirical Paper

This is a VERY VERY important distinction that you must learn and understand

Literature Review. What does it do???

Analyzes and compares *existing* research.

It uses all of the following

1. books
2. journal articles
3. published studies

It is clear that its goal is to

Understand what researchers already know.

No original data collection required.

Empirical Paper is way different

It collects and analyzes original data.

Examples could be

- surveys
- interviews
- experiments
- observations

**So basically you are trying to
produce new evidence or findings.**

Scope Control (SC) is important while doing research

Some of you are trying to research EVERYTHING.
I had one student ask me if that is a good RQ “The effects of technology on society”
Impossible to execute. Again Impossible. No need to explain.

Narrow By	Example
Time	2020–2025
Place	Uzbekistan
Population	Teenagers
Platform	TikTok
Variable	Political polarization

Better use this table to narrow down the scope

Also, please be realistic with what you are planning to do. Do not try to solve climate change or interview Putin and Trump. You are high school students so choose your RQ wisely

Scholarly Databases - Where can you get the resources? (Please not TikTok)

- Google Scholar
- JSTOR
- PubMed
- ScienceDirect
- Or you can simply google your topic and the papers will appear themselves. Make sure the source is credible

Search Strategies

1. Use Quotation Marks (" ")

Quotation marks search for the exact phrase

Example right here -

"political polarization"

Without quotation marks:

---> **Google Scholar searches the words separately**

With quotation marks:

-----> **it searches the exact term together**

2. Use AND

AND narrows your search by combining ideas

Example:

"political polarization" AND social media

Results must include:

political polarization

social media

This helps find sources specifically connecting both topics.

3. Filter by Year

You guys can use filters to find:

recent studies

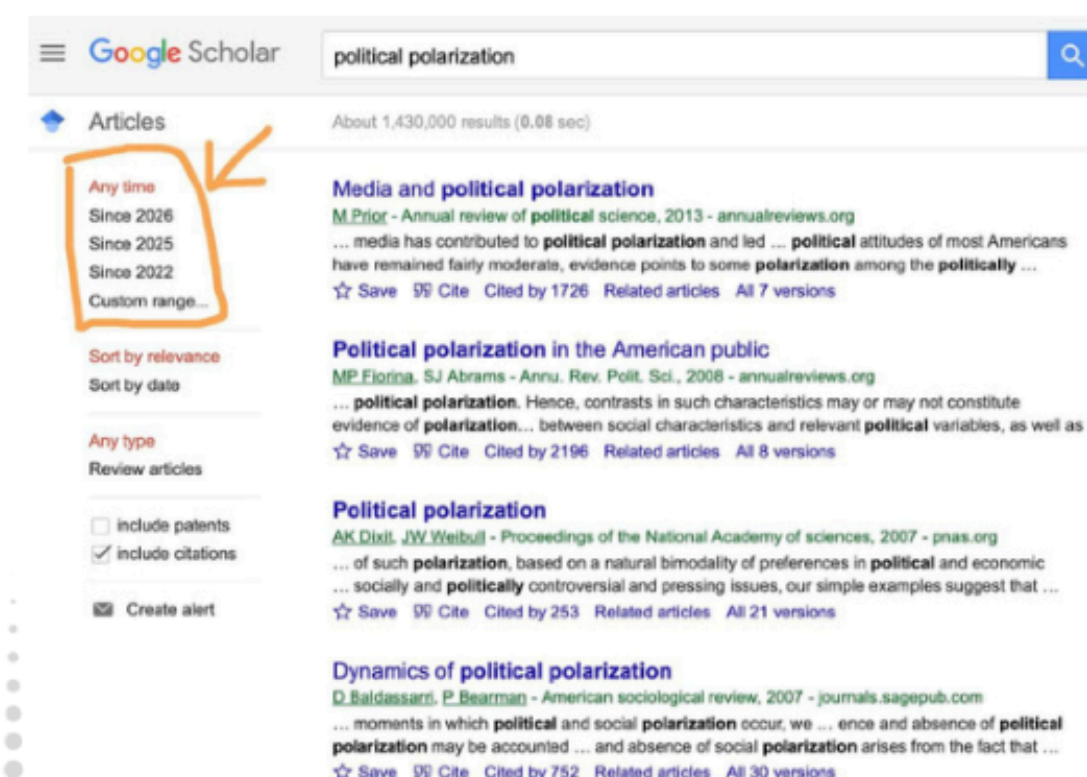
historical studies

Example:

Filter: 2021-2026

useful for fast-changing topics like technology or AI

Newer sources are often more relevant in modern research!!! Please try to use sources no later than 2020



We will use the simplified CRAAP Method

**How Do We Know a
Source is Reliable?**



Criteria	Question to Ask
Currency	Is the information recent enough?
Relevance	Does it actually relate to your topic?
Authority	Who wrote or published it?
Accuracy	Is the evidence trustworthy and supported?
Purpose	Why was this source created?

1. Currency

Ask:

"When was this published?"

Example:

AI article from 2012 → probably outdated

AI article from 2025 → more useful

Some topics change quickly, so newer sources matter more.

2. Relevance

Ask:

"Does this source directly help answer my research question?"

Example:

Topic:

Social media and political polarization

Weak source:

General history of television

Strong source: Study on TikTok algorithms and political opinions

3. Authority

Ask:

"Who is the author?"

Strong authority:

university professor

researcher

academic journal

government institution

Weak authority:

anonymous blog writer

random social media account

Example:

A study from Harvard

University usually has stronger

credibility than an unknown

blog.

4. Accuracy

Ask:

"Is the information supported by evidence?"

Look for:

citations

statistics

data

references

peer review

Warning signs:

no evidence

emotional language

unsupported claims

Example:

"Studies show..." → good if sources are provided

"Everyone knows..." → weak academic evidence

5. Purpose

Ask:

"Why was this source created?"

Possible purposes:

educate

inform

persuade

advertise

entertain

Example:

Research paper → usually informative

Company advertisement → may be biased

GUYS!

Every source has a perspective or goal.

You do not just go to school for fun. You go

there to get education, right? Or just talk

with friends ... (not good)

What is an Annotated Bibliography?

An annotated bibliography is a list of research sources where each citation is followed by a short explanation called an annotation.

Example --->

Max, Khamraev. How to Stop Me From Writing "lock in" in The Chat. Terra Academic Press, 2026.

Annotation

This book shows the emotional and physical struggles students experience while seeing the Terra Research Mentor pressuring them to focus on the work. This surely includes sleep deprivation, excessive caffeine consumption, and last-minute panic. The author is a well-known students' nightmare and former student at Hewlett High School. This source would be useful for researching why students stress when seeing the nickname "khamraevvv".

What Does an Annotation Include?

Each annotation usually contains 3 parts:

Part Purpose

Summary What is the source about?

Evaluation Is the source credible and useful?

Relevance How does it help your research topic?

Part 1 - Choose a research topic.

Select a topic you are genuinely interested in. It can be anything

Social media and mental health/AI/Education/Political science/Medicine/Climate change and etc.

Part 2 - Create a research question using the formula we gave you

Make sure the RQ is narrow enough and you are controlling the scope. RQ should be 10-15 words

Part 3 - Find Scholarly Sources

Using: Google Scholar, JSTOR or another academic database

You will need to find ---

15 academic/scholarly sources

Requirements: published within the last 10 years & related to your chosen topic.

Part 4 - do the CRAAP method for every single paper to prove credulity.

Part 5 - Create a Mini Annotated Bibliography

Include:

a short summary

evaluation of credibility

explanation of usefulness to your topic